Technical Design Note

Power Electronics and Control Systems

Three-phase dual active bridge (DAB3)

1200 V / 1200 V, 250 kW, 10 kHz: sizing, control, simulation model and runs

Isolated bidirectional DC/DC stage with two six-step bridges of SiC modules and three single-phase transformers in star-star, with the DC bias of the transformers and the first four runs of the Simscape bench.

Authors:	**Revision:**	**Date:**
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Abstract: The three-phase dual active bridge (DAB3) transfers power between two DC links through two three-phase bridges driven with six-step voltages and three single-phase transformers in star-star with floating star points; the power is set by the phase displacement between the two bridges. This note documents the 250 kW DAB3 of the folder `analysis_three_phase_dab` of `pwr-control/advanced-dcdc-converters`, between two batteries at about 1200 V, with the transformers at 10 kHz. The sizing is derived step by step on a harmonic model of the converter (operating point, series inductance and phase shift, transformer and choice of the frequency, SiC modules and zero-voltage switching, DC links, loss budget), and the DAB3 is compared with the single-phase DAB at the same point. The current control, with its feed-forward, is designed on the LC of the DC side, and its sample rate is discussed for a control that runs at the transformer frequency or slower. A section is given to the DC bias of the transformers: where it comes from, what limits it, how to start the converter and how to update the phase shift without creating it, and how to measure and cancel it when the current is sampled no faster than the switching frequency. The second part describes the Simscape bench, built with the blocks of `pwr-control/library` and a parameter structure of its own, the changes made to the original model (which ran the transformers at 2 kHz), the numerical settings and their artefacts, and the first four runs: forward and reverse power at 1 pu, light load, and voltage mismatch. The runs agree with the harmonic model within 2 % on the currents and the switching currents and confirm the map of the soft-switching range.



Revision history

Rev.	Date	Description	Authors
00	October 2026	First issue: sizing, control, DC bias of the transformers, simulation bench and runs 01 to 04.	Davide Bagnara

Project Repository: advanced-dcdc-converters.git
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Nomenclature

Symbol	Meaning
V_1, V_2	DC voltages of side 1 and side 2, 1200 V nominal
$n = n_1/n_2$	turns ratio of the three transformers, 14 : 14
$V_2' = nV_2$	voltage of side 2 referred to side 1
$d = V_2'/V_1$	voltage ratio
P, P_{nom}	power from side 1 to side 2; nominal power, 250 kW in both directions
f_{s}, ω, T_{s}	transformer frequency, which is also the switching frequency of every device, 10 kHz; $\omega = 2\pi f_{\mathrm{s}}$; $T_{\mathrm{s}} = 1/f_{\mathrm{s}}$
f_{PWM}	clock of the library modulator, one step of the six-step pattern per period: $f_{\mathrm{PWM}} = 6f_{\mathrm{s}} = 60\,\mathrm{kHz}$
$\theta = \omega t$	angle within the period, $\theta = 0$ at the rising edge of leg a of bridge 1
φ	phase displacement of bridge 2 after bridge 1; the power flows from side 1 to side 2 for $\varphi > 0$
φ_{nom}, φ_{max}	phase shift at nominal power (30°); limit of the modulator (60°)
u	output of the current control, $u \in [-1, 1]$, $\varphi = u\,\varphi_{\mathrm{max}}$
L	series inductance of a phase, referred to side 1: $L = L_{\mathrm{d1}} + L_{\mathrm{d2}}'$
L_{d1}, L_{d2}	leakage plus external inductance of the primary and of the secondary
L_{m}, R_{fe}	magnetizing inductance and core-loss resistance, primary side
R_{s1}, R_{s2}	resistance of winding, terminations and external inductor, primary and secondary
R_1, R_2	series resistance of a phase: R_{s1} or R_{s2} plus the on-resistance of the switch
$u_{1\mathrm{x}}$, $u_{2\mathrm{x}}$	phase-to-star-point voltages of bridge 1 and bridge 2, $x \in \{a, b, c\}$
$i_{1\mathrm{x}}$, $i_{2\mathrm{x}}$	phase currents, out of leg x of bridge 1 and into leg x of bridge 2
i_{m}	magnetizing current with the core-loss current, primary side
i_{dc1}, i_{dc2}	DC currents of the bridges: into bridge 1 from DC link 1, out of bridge 2 into DC link 2
I_{dc1}	DC input current of side 1, the current of the DC choke where it enters the DC link (the controlled quantity)
i_{sw1}, i_{sw2}	switching currents of the two bridges, positive when they charge the leg capacitances in the direction of zero-voltage switching
C_{oss}	output capacitance of a switch



Symbol	Meaning
t_d	dead time, 500 ns
t_c	step of the Simscape local solver, 100 ns in the runs
λ, B	flux linkage of a winding and flux density of its core
N, A_c	turns and core area of a transformer
C_dc, L_dc	DC-link capacitor and DC choke of each side
k_ff	gain of the feed-forward of the current control
k_p, k_i	gains of the PI of the current control
h	harmonic order, $h = 6k \pm 1$
pu	per unit; the current base is $P_\mathrm{nom}/V_1 = 208\,\mathrm{A}$
ZVS	zero-voltage switching
SoC	state of charge of a battery
BE	Backward Euler integration method

1. Introduction and scope

The solid state transformer study of [7] compared three isolated DC/DC stages at about 250 kW per phase: the three-phase dual active bridge of De Doncker, Divan and Kheraluwala [1] (DAB3) at 4 kHz on three single-phase transformers of nanocrystalline cut cores, the single-phase DAB and the CLLC, the stage that the SST bench of today uses [8]. The study left the DAB3 to the repository `advanced-dcdc-converters` [9]. This note is that continuation: it documents the folder `analysis_three_phase_dab` of the repository, turned into a 250 kW isolated bidirectional DC/DC stage between two batteries.

Figure 1 is the converter. Each side has a battery behind a DC choke and a DC-link capacitor. The two three-phase bridges of 2 kV SiC MOSFET half-bridge modules make six-step voltages at the transformer frequency $f_\mathrm{s} = 10\,\mathrm{kHz}$. Three single-phase transformers, one per phase, are connected in star on both sides with the star points floating; their leakage, completed by an external inductor, is the series inductance that carries the power. The only control variable is the phase displacement φ of bridge 2 after bridge 1, set by a loop on the DC input current of side 1.

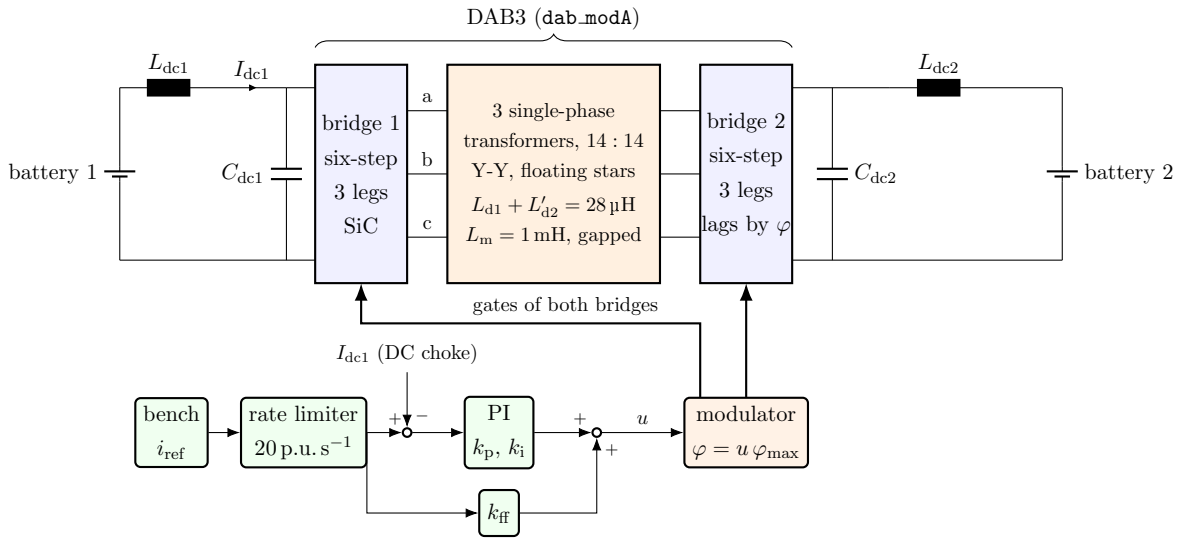


Figure 1: The DAB3 between two batteries, as in the bench of Section 6: DC chokes and DC links on both sides, two six-step bridges, three single-phase transformers in star-star with floating star points. The current loop regulates the DC input current of side 1, I_{dc1}, with a PI and a feed-forward of the reference; the modulator turns its output u into the phase displacement φ of bridge 2 after bridge 1.

The work behind this note had four steps, each documented here. First, the initialization of the model was rewritten around a parameter structure of its own, with the numbers of this converter in `private/dab3_param.m`, independent of the data of the library [10] while still using its blocks and Simscape components. Second, the original model was reviewed: it ran the transformers at 2 kHz without showing it, its modulator limit was beyond the maximum of the power, and its DC links, chokes and batteries came from the data of the single-phase DAB (Section 6.2). Third, the converter was sized at 1200 V / 1200 V and 10 kHz on a harmonic model (Sections 2 to 4). Fourth, four runs of the Simscape bench were checked against the harmonic model (Section 7).

Working assumptions. The DC voltages (1200 V on both sides; the batteries of the bench stay between about 1180 V and 1270 V), the turns ratio 1 : 1 and the transformer frequency of 10 kHz are assumptions, to be confirmed with the specification. With other values the procedure is the same and the numbers of Section 3 change. The control of the hardware is expected to run at 10 kHz at most, that is at most once per period of the transformers: Sections 4.3 and 5 take this into account.

2. Operating principle

2.1. Six-step bridges and star-star transformers

Each leg of a bridge is switched at 50 % duty and the three legs are displaced by 120°. With the star points floating, the phase-to-star-point voltage of each transformer takes the levels $\pm V/3$ and $\pm 2V/3$, each for a sixth of the period (Figure 3), and contains only the harmonics $h = 6k \pm 1$:

$$u_{1a}(\theta) = \frac{2V_1}{\pi} \sum_{h=1,5,7,11,\ldots} \frac{\sin h\theta}{h}, \qquad u'_{2a}(\theta) = \frac{2V'_2}{\pi} \sum_{h=1,5,7,11,\ldots} \frac{\sin h(\theta-\varphi)}{h}, \tag{2.1}$$

with leg a of bridge 1 rising at $\theta = 0$ and legs b and c 120° and 240° later. The fundamental has the amplitude $2V/\pi$, 764 V at 1200 V (936 V line to line, rms); the triplen harmonics, which would drive a zero-sequence current, cannot appear with floating star points. Each switch turns on and off once per period: the switching frequency of every device is the transformer frequency.

2.2. Power transfer

With the converter lossless and the magnetizing inductance neglected, the series inductance $L = L_{d1} + L'_{d2}$ of each phase carries the power [1]

$$P(\varphi) = \frac{V_1 V'_2}{\omega L} \cdot \begin{cases} \varphi\left(\dfrac{2}{3} - \dfrac{|\varphi|}{2\pi}\right), & |\varphi| \le \dfrac{\pi}{3}, \\ \operatorname{sign}(\varphi)\left(|\varphi| - \dfrac{\varphi^2}{\pi} - \dfrac{\pi}{18}\right), & \dfrac{\pi}{3} \le |\varphi| \le \dfrac{2\pi}{3}, \end{cases} \tag{2.2}$$

which grows up to $|\varphi| = \pi/2$, where $P_{max} = \frac{7\pi}{36} V_1 V'_2/(\omega L)$, and falls beyond it (Figure 2). The control drives the library modulator with $u \in [-1, 1]$ and the modulator applies $\varphi = u\,\varphi_{max}$. The original model had $\varphi_{max} = 2\pi/3$: the maximum of the power was inside the range of u, and the gain of the plant changed sign at $u = 0.75$. With $\varphi_{max} = \pi/3$ the power is monotonic in u and reaches 429 kW ($1.7\,P_{nom}$) at nominal voltages and 364 kW ($1.45\,P_{nom}$) with $d = 0.85$.

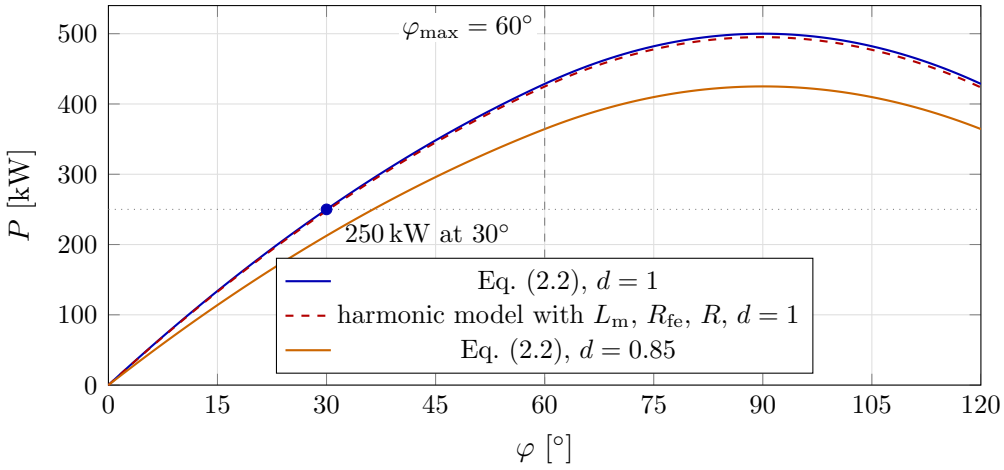


Figure 2: Power against the phase shift at 1200 V, $L = 28\,\mu H$, 10 kHz (`figures/power_phi.dat`): Eq. (2.2) at $d = 1$ and $d = 0.85$, and the harmonic model of Section 2.3 with the magnetizing branch and the resistances (power delivered to side 2). The modulator limit $\varphi_{max} = 60°$ keeps the control on the rising part of the curve.

2.3. Harmonic model

The sizing and the checks of the runs use a steady-state model by harmonic superposition: `private/dab3_harmonic_op.m` in the repository, and the same model in `py/three_phase_dab_figures.py` for the figures of this note. Per phase, the transformer is the T equivalent referred to side 1: $Z_{1h} = R_1 + jh\omega L_{d1}$ on side 1, the magnetizing branch $Z_{mh} = jh\omega L_m \parallel R_{fe}$, and $Z_{2h} = R'_2 + jh\omega L'_{d2}$ on side 2, where R_1 and R_2 include the on-resistance of the switches. For each harmonic of Eq. (2.1), with $U_{1h} = 2V_1/(h\pi)$ and $U_{2h} = 2V'_2/(h\pi)\, e^{-jh\varphi}$,

$$E_h = \frac{U_{1h}/Z_{1h} + U_{2h}/Z_{2h}}{1/Z_{1h} + 1/Z_{mh} + 1/Z_{2h}}, \qquad I_{1h} = \frac{U_{1h} - E_h}{Z_{1h}}, \qquad J_{2h} = \frac{E_h - U_{2h}}{Z_{2h}}, \tag{2.3}$$

and the currents of phase a are $i_{1a}(\theta) = \sum_h \mathrm{Im}\{I_{1h} e^{jh\theta}\}$ and $i_{2a}(\theta) = n \sum_h \mathrm{Im}\{J_{2h} e^{jh\theta}\}$, the other phases 120° and 240° later. The quantities of the sizing follow from the phasors:

$$P_1 = \frac{3}{2} \sum_h \mathrm{Re}\{U_{1h} I_{1h}^*\}, \qquad \begin{aligned} i_{sw1} &= -i_{1a}(0) = -\sum_h \mathrm{Im}\{I_{1h}\}, \\ i_{sw2} &= i_{2a}(\varphi) = n \sum_h \mathrm{Im}\{J_{2h}\, e^{jh\varphi}\}. \end{aligned} \tag{2.4}$$

i_{sw1} is the current that leg a of bridge 1 commutates at its rising edge, positive when it flows into the leg and charges the node upwards as ZVS requires; i_{sw2} is the same for bridge 2, whose current is counted into the leg. The falling edges see the same currents with the opposite sign, by half-wave symmetry. In the lossless case the model agrees

with Eq. (2.2) within 0.05 %. With the magnetizing branch and the resistances it gives the currents, the switching currents, the magnetizing current and the ripple of the DC side used in the following sections, and each run of Section 7 is compared with it at the voltages and the phase shift measured in the run.

2.4. Waveforms at the nominal point

Figure 3 shows one period at 1200 V / 1200 V and $\varphi = 30°$ with the numbers of Section 3. The phase currents are 161 A rms and 240 A peak. At its rising edge each leg of bridge 1 commutates 124 A flowing into the leg, each leg of bridge 2 125 A flowing from the transformer into the leg: both bridges switch with ZVS (Section 3.4). The magnetizing current follows the flux of the core, 11.7 A peak. The DC current of each bridge is the sum of the currents of the legs whose upper switch is on: it has the six-pulse shape at $6f_s = 60$ kHz, between 115 A and 240 A around its mean of 207 A, that is 40 A rms of ripple for the DC-link capacitors.

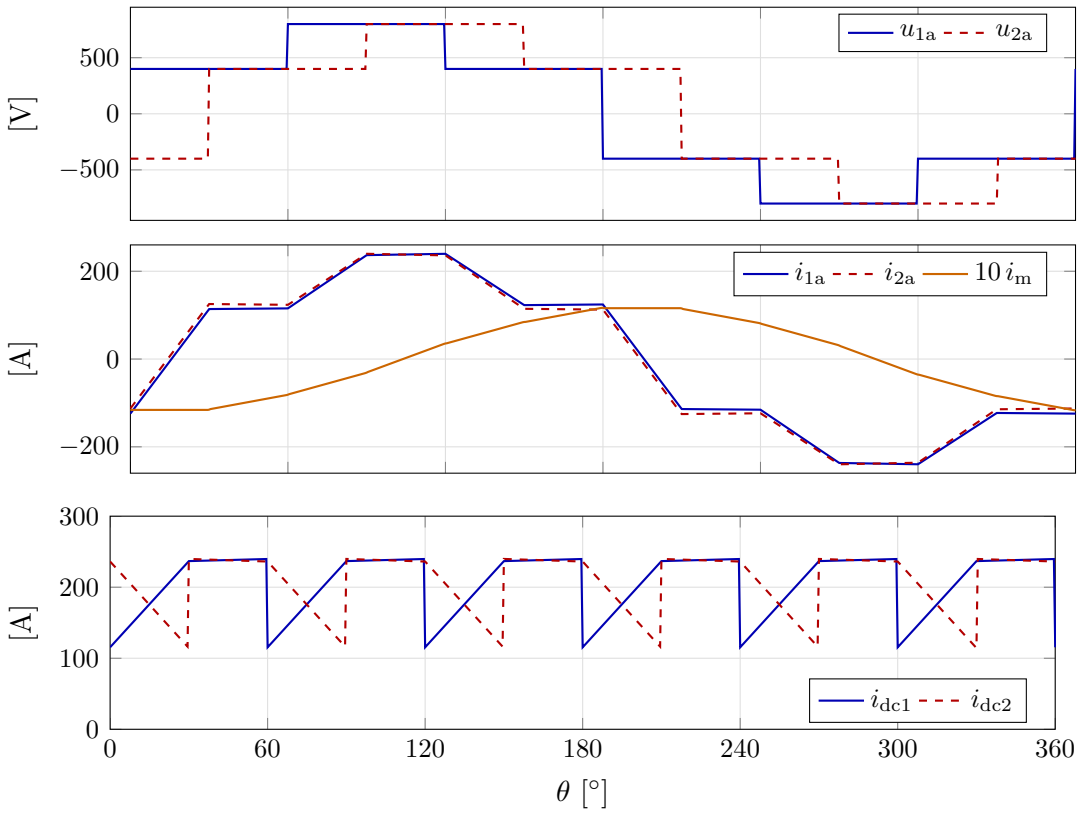


Figure 3: Steady state at 1200 V / 1200 V, $\varphi = 30°$, from the harmonic model (`figures/waveforms_nominal.dat`): phase-to-star-point voltages of the two bridges (top), phase currents of phase a on both sides and magnetizing current, scaled by 10 (middle), DC currents of the two bridges (bottom).

2.5. Why three phases

Table 1 compares the DAB3 with a single-phase DAB [2, 3] at the same point, both lossless and with the magnetizing inductance neglected (`py/three_phase_dab_figures.py`). For the same power and phase shift the DAB3 needs less series inductance per phase, its phase current is two thirds of the current of the single-phase converter, it turns off less than half the current at each commutation, and the ripple of its DC current is a third in amplitude at three times the frequency, so that with the same 500 µF the voltage ripple of the DC link is six times smaller. The price is the number of switches, 12 against 8, and of magnetic components, three transformers with a third of the power each. With the same device in every position the conduction losses are 30 % lower and the sum of the turned-off currents 29 % lower; with the same total silicon area the conduction losses are about the same, and the advantage stays in the switching losses, in the DC links and in the volt-seconds per transformer. The 2025 comparison of [7] found the single-phase DAB more efficient, but at different operating points (the single-phase DAB at 12 kHz, the DAB3 at 4 kHz, with different transformers), and it left the residual DC current of the single-phase transformer as an open question; Section 5 treats that question for the DAB3.

Table 1: DAB3 and single-phase DAB at 250 kW, 1200 V / 1200 V, 10 kHz, $\varphi = 30°$, lossless, L_m infinite (`py/three_phase_dab_figures.py`).

Quantity	**DAB3**	**Single-phase DAB**
Series inductance	28.0 µH per phase	40.0 µH
Current of a winding	161 A rms, 238 A peak	236 A rms, 250 A peak
Current at the commutations	119 A	250 A
Switches; conduction with 1.04 mΩ each	12; 162 W	8; 231 W
Sum of the turned-off currents per period	1428 A (12 × 119 A)	2000 A (8 × 250 A)
Ripple current of the DC link	38.5 A rms at 60 kHz	110.5 A rms at 20 kHz
Voltage ripple with 500 µF	0.56 V peak to peak	3.5 V peak to peak
Transformers	3, 13.3 mV s each, 274 kV A of windings per side	1, 30.0 mV s, 283 kV A



3. Sizing of the 250 kW stage

3.1. Operating point

Table 2 is the operating point. Both DC voltages are 1200 V, the voltage of the 690 V application case of the library data used by the original model, so the transformers are 1 : 1 and the same SiC module is used on both bridges. The base current of the control is $P_{\mathrm{nom}}/V_1 = 208\,\mathrm{A}$.

Table 2: Operating point of the DAB3.

Quantity	Value	Note
DC voltages V_1, V_2	1200 V / 1200 V	assumption; batteries of the bench 1248 V at SoC 0.85
Turns ratio	1 (14 : 14)	three single-phase transformers, Y-Y, floating star points
Nominal power	250 kW	both directions
DC current	208 A	base of the current loop
Transformer frequency f_{s}	10 kHz	switching frequency of every device
Clock of the modulator f_{PWM}	60 kHz $= 6f_{\mathrm{s}}$	one step of the six-step pattern per period of the clock
Series inductance L	28 µH per phase	14 µH + 14 µH, leakage and external inductor
Phase shift at 250 kW	30°	Eq. (2.2)
Limit of the modulator $\varphi_{\max}$	60°	429 kW at $d = 1$, 364 kW at $d = 0.85$
Devices	`danfoss_SKM1700MB20R4S2I4`	2 kV SiC half-bridge modules of the library datasets, 1.04 mΩ, $C_{\mathrm{oss}} = 3.28\,\mathrm{nF}$
Dead time t_{d}	500 ns	to be confirmed; not applied by the library modulator
Current control	PI and feed-forward on I_{dc1}	18 Hz of bandwidth, Section 4

3.2. Series inductance and phase shift

The phase shift chosen for the nominal point sets the series inductance through Eq. (2.2):

$$L = \frac{V_1 V_2'}{\omega P_{\mathrm{nom}}}\,\varphi_{\mathrm{nom}}\left(\frac{2}{3} - \frac{\varphi_{\mathrm{nom}}}{2\pi}\right) = 28.0\,\mu\mathrm{H} \quad (\varphi_{\mathrm{nom}} = 30°). \tag{3.1}$$



Table 3 shows the trade-off at 10 kHz. The rms current changes little with φ_{nom}. A small φ_{nom} gives a large power gain $\mathrm{d}P/\mathrm{d}\varphi$, which makes the converter sensitive to the timing of the gates and to the resolution of the phase shift; a large one leaves less margin for the voltage range and for the transients. 30° is the middle choice: 28 µH per phase, $P_{\max} = 2P_{\mathrm{nom}}$ at 90° and $1.7P_{\mathrm{nom}}$ at $\varphi_{\max}$. The library class `hw_dab_three_phase_setup` derives its own series inductance from the PWM frequency with the single-phase expression $V_1^2/(4f_{\mathrm{PWM}}P)$, 24 µH with the inputs of this converter; `init_model.m` writes the 14 µH + 14 µH of the sizing over it. The leakage of the transformers is estimated at about 12 µH in total (Section 3.3); the rest is an external or integrated inductor.

Table 3: Series inductance against the nominal phase shift, 250 kW, 1200 V / 1200 V, 10 kHz; currents from the harmonic model with $L_{\mathrm{m}} = 1$ mH.

φ_{nom}	L per phase	Phase current, rms / peak	i_{sw}	$P_{\max}$ at $d = 1$ / $d = 0.85$
20°	19.6 µH	156 A / 229 A	119 A	716 kW / 609 kW
25°	23.9 µH	159 A / 234 A	122 A	586 kW / 498 kW
30°	**28.0 µH**	**161 A / 240 A**	**124 A**	**500 kW / 425 kW**
35°	31.9 µH	164 A / 245 A	127 A	439 kW / 373 kW
40°	35.6 µH	167 A / 251 A	130 A	394 kW / 335 kW

3.3. Transformer

Each phase has its own single-phase transformer on the core set of the 2025 study [7]: three AM-NC-320C nanocrystalline cut cores in parallel, $A_{\mathrm{c}} = 47\,\mathrm{cm}^2$, 26.35 kg. The phase-to-star voltage of the six-step pattern has the peak flux linkage

$$\lambda_{\mathrm{pk}} = \frac{V_1}{9 f_{\mathrm{s}}} = 13.3\,\mathrm{mV\,s}, \tag{3.2}$$

half of the integral of $V/3$, $2V/3$, $V/3$ over three sixths of the period; at full load the drop on L_{d1} lowers it to 11.6 mV s. With 14 turns the peak flux density is 0.20 T at no load and 0.18 T at full load (Table 4).

The magnetizing inductance is a choice. The gapped core with $L_{\mathrm{m}} = 1$ mH costs a magnetizing current of about 12 A peak, which adds to the switching currents and keeps both bridges in ZVS down to 9 % of the load at $d = 1$ (Section 3.4); it also limits the flux offset that a DC current produces (Section 5). With the relative permeability of 5000 of the cut cores [7] and no gap, L_{m} would be about 8 mH: the ZVS limit at $d = 1$ would move from 9 % to about 15 % of the load, and the core would saturate with less than 10 A of DC.



Table 4: Transformers, one per phase (values marked TBC to be confirmed with the design of the windings).

Quantity	**Value**	**Note**
Core	3 × AM-NC-320C, nanocrystalline cut cores	$A_c = 47\,\mathrm{cm}^2$, 26.35 kg, the set of [7]
Peak flux linkage	13.3 mV s	Eq. (3.2); 11.6 mV s at full load
Turns	14 : 14	0.20 T peak at no load, 0.18 T at full load
Magnetizing inductance	1 mH, gapped	i_m 12 A to 13 A peak; $\mu_{r,eff} = 622$ with $l_e = 0.72\,\mathrm{m}$ (TBC)
Core losses	about 107 W	$R_{fe} = 3\,\mathrm{k\Omega}$; $16\,\mathrm{W\,kg^{-1}}$ at 4 kHz and 0.8 T [7], scaled with $f^{1.5}B^2$
Windings, terminations, external inductor	$R_{s1} = R_{s2} = 1.3\,\mathrm{m\Omega}$ (TBC)	68 W per transformer at 161 A rms
Leakage	about 12 µH in total	with 10 mm between the windings; the rest of the 28 µH is external

Choice of the frequency. The transformer frequency is the main choice of the design. Table 5 compares four frequencies at the nominal point with the same core set, the turns set for $B \leq 0.35\,\mathrm{T}$, the same devices, L scaled for 30° at 250 kW, and L_m scaled to keep the same magnetizing current; the turn-off losses are the upper bound of Section 3.6. The losses about double with each doubling of the frequency, driven by the turn-off and the core terms, while the transformer at 2 kHz needs 41 turns for five times the volt-seconds of 10 kHz. 10 kHz keeps the losses near 1 % and the magnetics compact; 5 kHz is the conservative alternative. At 10 kHz the final design uses 14 turns instead of the 9 of the comparison, which lowers the flux density to 0.20 T and the core losses from 776 W to 320 W. The original model ran at 2 kHz without showing it: its PWM frequency was 12 kHz and the library modulator makes the six-step pattern at $f_{PWM}/6$.

Table 5: Losses at 250 kW against the transformer frequency: same core set with the turns for $B \leq 0.35\,\mathrm{T}$, same devices, 30° at the nominal point. Losses in watts; the total includes the body diodes in the dead time (5 to 52 W).

f_s	L	N	B	Conduction	Turn-off	Windings	Cores	**Total**
2 kHz	140 µH	41	0.35 T	162	299	195	84	**0.74** kW
5 kHz	56 µH	17	0.33 T	162	746	195	308	**1.42** kW
10 kHz	28 µH	9	0.32 T	162	1493	195	776	**2.65** kW
20 kHz	14 µH	5	0.28 T	162	2985	195	1778	**5.17** kW

3.4. Semiconductors and zero-voltage switching

Both bridges use the 2 kV SiC half-bridge module of the original model, the dataset `danfoss_SKM1700MB20R4S2I4` of the library: 1.04 mΩ, C_{oss} = 3.28 nF, E_{on} and E_{off} of 77 mJ and 108 mJ at 1300 V and 1000 A. The module is oversized for 161 A rms (162 W of conduction for the 12 switches): a module of the 300 A to 600 A class would do, with a smaller C_{oss} and a wider ZVS range.

At each commutation the current of the phase charges one C_{oss} of the leg and discharges the other. The transition is complete within the dead time when

$$i_{sw} \geq \frac{2C_{oss}V}{t_d} = 15.7\,\text{A} \qquad (V = 1200\,\text{V},\ t_d = 500\,\text{ns}), \tag{3.3}$$

and it lasts $2C_{oss}V/i_{sw}$. At the nominal point i_{sw} = 124 A on both bridges and the transition takes 63 ns. With $d \neq 1$ the bridge of the lower voltage loses its switching current at light load: Figure 4 and Table 6 give, from the harmonic model, the lowest load above which each bridge has the current of Eq. (3.3). At $d = 1$ the magnetizing current keeps both bridges in ZVS down to 9 % of the load; a mismatch of 5 % moves the limit to about 30 %, a mismatch of 15 % to 59 or 72 %. This is the known limit of the DAB3 with six-step voltages and phase shift only [1]. If the batteries can differ by more than about 5 %, the options are a DC link that follows the battery (a front-end stage), a lower L_m, or a modulation with a third degree of freedom, such as pulse width or asymmetric pulses.

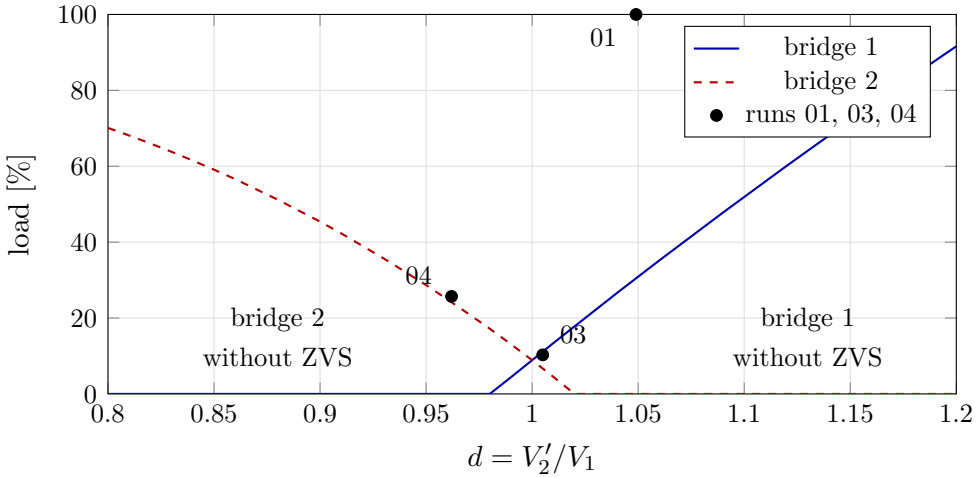


Figure 4: Lowest load at which each bridge commutates the current of Eq. (3.3) within the dead time of 500 ns, forward power, V_1 = 1200 V, harmonic model with L_m = 1 mH (`figures/zvs_map.dat`); ZVS above the lines. The points are the forward runs of Section 7 at their measured d and load. In reverse power the roles of the bridges are exchanged and d is replaced by $1/d$.



Table 6: Lowest load with ZVS of each bridge against the voltage ratio (Figure 4).

d	0.85	0.90	0.95	1.00	1.05	1.10	1.15
Bridge 1	0	0	0	9 %	31 %	52 %	72 %
Bridge 2	59 %	45 %	29 %	9 %	0	0	0

The model does not simulate the transitions. The library modulator does not apply its dead-time parameter and the bridges have no C_{oss}: the legs switch at once. The runs give the switching currents, with their sign and value at the edges, and the transition times are computed from them with Eq. (3.3).

3.5. DC links and chokes

The DC current of each bridge has the six-pulse ripple of Figure 3, 40 A rms at 60 kHz at full load, which gives 0.6 V peak to peak on 500 µF: the DC links are film banks sized on the ripple current rather than on the voltage ripple. Between the batteries and the DC links there are chokes of 250 µH and 2 mΩ. With the DC link they resonate at 450 Hz, damped by the internal resistance of the battery (72 mΩ with the settings of the bench), and this resonance limits the bandwidth of the current loop (Section 4). The DC links and the batteries start at the open-circuit voltage of the batteries. The values of the DC side are those of the bench and are still to be confirmed.

3.6. Loss budget

Table 7 is the estimate at 250 kW. The turn-off term scales the E_{off} of the dataset linearly with voltage and current to 124 A. It is an upper bound: with ZVS the C_{oss} acts as a snubber and the turn-off energy is a fraction of the hard-switching one. The soft turn-off energy of the module at 1200 V and 100 A to 150 A is the missing number: with it the efficiency is between 99.0 % and 99.6 %.



Table 7: Loss budget at 250 kW (sizing estimate).

Item	Loss	Basis
Conduction, 12 switches	162 W	1.04 mΩ, 161 A rms per leg
Turn-off, 12 switches	≤ 1490 W	E_{off} of the dataset scaled with VI at 124 A: upper bound
Turn-on	0	ZVS
Body diodes in the dead time	26 W	500 ns minus 63 ns at 124 A, 4 V
Windings, terminations, external inductors	203 W	1.3 mΩ per side and phase
Cores	320 W	$R_{\text{fe}} = 3\,\text{k}\Omega$
DC chokes and DC links of the bench	about 175 W	2 mΩ per side at 208 A, 1 mΩ at 40 A rms
Total	**about 2.4** kW	$\eta \simeq 99.0\,\%$; 99.6 % if the turn-off losses are small

4. Current control

4.1. Structure

The control of the bench (subsystem `dab_ctrl` of the model, Figure 1) regulates the DC input current of side 1, I_{dc1}, normalized by 208 A. I_{dc1} is the current of the DC choke where it enters the DC link (`dab_current_input_modA`): it has no six-pulse ripple, which stays in the DC-link capacitor. This is the feedback of the original model. The reference of the bench goes through a rate limiter of $20\,\text{p.u.}\,\text{s}^{-1}$; the PI of the library (`pi_ctrl`) adds its output to the feed-forward $u_{\text{ff}} = k_{\text{ff}}\, i_{\text{ref}}$, and the modulator applies $\varphi = u\,\varphi_{\text{max}}$. With $k_{\text{ff}} = \varphi_{\text{nom}}/\varphi_{\text{max}} = 0.5$ the feed-forward gives 30° at 1 pu: exact at the nominal point, about 1° too much at half load, because Eq. (2.2) is not linear. The voltage loop of the original model (`Vout Ctrl`) is kept in the model but not selected.

4.2. Design

The plant gain $\text{d}(I_{\text{dc1}}/208\,\text{A})/\text{d}u$ is about 1.9 at the operating point of the bench (about 1220 V / 1280 V, 28°) and 1.7 at 1200 V / 1200 V. The LC of the DC side puts a lightly



damped pole pair at 450 Hz to 500 Hz in the loop and sets its bandwidth. Table 8 gives the margins of the discrete loop (PI, one sample of delay on the feedback, zero-order hold, battery impedance) for four sets of gains: the gain margin, at the resonance, is the one that runs out. The chosen $k_\mathrm{p} = 0.01$ and $k_\mathrm{i} = 60$ give 18 Hz of crossover, 90° of phase margin and 14 dB of gain margin. The bandwidth is low on purpose: the feed-forward does the tracking and the integrator removes the static error. On an averaged model of the bench (batteries, chokes, DC links, Eq. (2.2), quantization of φ) the loop follows the ramps of 20 p.u. s^{-1} within 0.024 pu and has no static error 10 ms after the end of a ramp. With the gains of the original model ($k_\mathrm{p} = 0.02$, $k_\mathrm{i} = 1$, $k_\mathrm{ff} = 0.25$ with $\varphi_\mathrm{max} = 2\pi/3$) the error of the feed-forward, 5 % at 1 pu on this bench, lasted about one second.

Table 8: Margins of the discrete current loop at the operating point of the bench.

k_p, k_i	Crossover	Phase margin	Gain margin (at 450 Hz to 500 Hz)
0.01, 60 (chosen)	**18** Hz	**90°**	**14** dB
0.02, 100	30 Hz	91°	10 dB
0.05, 150	45 Hz	94°	6.5 dB
0.10, 300	95 Hz	98°	0.4 dB

4.3. Sample rate

In the model the PI is triggered by the modulator once per period of its clock, that is once per step of the six-step pattern, at 60 kHz. The control of the hardware will run at 10 kHz at most, once per period of the transformers. This does not limit the current loop, whose bandwidth is set by the LC of the DC side: one update per period adds about 125 µs of delay (sample and hold included), 0.8° at the crossover of 18 Hz and about 20° at the resonance, where the gain margin of Table 8 has to be checked again. Two points need care.

- **Feedback.** The six-pulse ripple of the DC current of a bridge is at 60 kHz, an integer multiple of a sampling at 10 kHz: sampled at a fixed angle it folds onto DC, and the measured current is wrong by an amount between −92 A and 33 A at full load, depending on the angle (Figure 3). The feedback has to be the current of the DC choke, as in the model, or an average over whole periods.
- **Update of the phase shift.** With one update per period, each leg would see the increment of φ first on its rising or first on its falling edge, and a ramp of φ would apply a DC voltage to the windings: Section 5.4 gives the rule that avoids it.



4.4. Reverse power

The loop regulates the current of side 1. In reverse power, at $-1\,\mathrm{pu}$ with battery 1 charging at 1271 V (run 02, Section 7), side 1 receives 265 kW and side 2 delivers 266.5 kW (220.6 A), 6 % above the nominal power. A power reference, or the current reference scaled by $1200\,\mathrm{V}/V_1$, keeps 250 kW in both directions.

5. DC bias of the transformers

5.1. Where it comes from

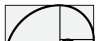
A DC component of the phase currents is driven by a DC component of the phase voltages, and the only thing that opposes it is the resistance of the path: at DC the magnetizing inductance is a short circuit, so a DC voltage on the phases of bridge 1 drives a current through $R_1 = R_{\mathrm{s1}} + R_{\mathrm{ds,on}} = 2.34\,\mathrm{m\Omega}$ and L_{m} only. The DC voltages come from anything that makes the volt-seconds of a leg differ between its high and its low state: different turn-on and turn-off delays of the gate drivers, different on-resistances of the upper and lower switches, asymmetric transitions. A leg that stays high 10 ns longer than half the period, at 1200 V and 10 kHz, has a DC voltage of 0.12 V with respect to the midpoint of the DC link, and its phase 0.08 V (the floating star point takes a third), that is

$$I_{\mathrm{dc}} = \frac{0.08\,\mathrm{V}}{2.34\,\mathrm{m\Omega}} \simeq 34\,\mathrm{A}, \qquad B_{\mathrm{dc}} = \frac{L_{\mathrm{m}} I_{\mathrm{dc}}}{N A_{\mathrm{c}}} = 0.52\,\mathrm{T}, \tag{5.1}$$

the second with the gapped $L_{\mathrm{m}} = 1\,\mathrm{mH}$ of the sizing. In steady state the DC current flows in the windings of side 1 and in the switches of bridge 1 and magnetizes the core: the transformer does not pass DC to side 2, and in the core the current is a flux offset. With the gap, 0.52 T added to the 0.2 T of the AC flux still stays below the saturation of the nanocrystalline material, but the core losses grow and the DC current shifts the switching currents of the rising and of the falling edges by $\pm 34\,\mathrm{A}$, which at light load means hard switching on one edge. Without the gap (L_{m} about 8 mH) the same current would saturate the core. A DC bias builds up with the time constant

$$\tau_{\mathrm{dc}} = \frac{L_{\mathrm{m}}}{R_1 \parallel R_2'} \simeq 0.85\,\mathrm{s}, \tag{5.2}$$

because the magnetizing branch discharges through both sides; the same time constant governs the decay of the offset left by the start of the converter (Section 5.3). The flux offset is therefore a slow phenomenon, and the measurement and the control that cancel it can be slow too. This was the open question of the single-phase DAB in [7]; it applies to the DAB3 in the same way, and it is treated in the literature both on the side of



the transients [4] and on the side of the steady state [5]. The note [6] treats the same problem for gapless single-phase transformers fed by an open-loop inverter, with the DC error sources of the inverter, the damping of the dead time in hard switching and a compensation on the $\alpha\beta$ components of the voltage; Section 5.6 follows it.

5.2. What limits it

Two things limit the DC bias without any control. The first is the gap of the cores, which turns a given DC current into a smaller flux offset, Eq. (5.1). The second is the soft switching itself. With ZVS the edge of a leg happens during the dead time, at the speed set by the switching current: a positive DC offset in phase a reduces the current at the rising edge of leg a and increases it at the falling edge, so the rising edge becomes slower and the falling edge faster, and both shorten the high time of the leg. The leg loses volt-seconds in proportion to the offset, as if a resistance

$$R_{\mathrm{zvs}} = \frac{2 C_{\mathrm{oss}} V^2 f_{\mathrm{s}}}{i_{\mathrm{sw}}^2} \tag{5.3}$$

were in series with the phase: 6.1 mΩ at full load ($i_{\mathrm{sw}} = 124\,\mathrm{A}$), more than twice the physical resistance, and 0.37 Ω at the ZVS limit ($i_{\mathrm{sw}} = 16\,\mathrm{A}$). With ZVS the 10 ns of Eq. (5.1) give about 9 A at full load and a fraction of an ampere at light load. The effect disappears with hard switching. It also damps the offset left by the start: with R_{zvs} added to R_1 and R_2', the time constant of Eq. (5.2) drops from 0.85 s to about 0.24 s at full load. The bench, without dead time and C_{oss}, does not show it.

5.3. Start-up

When the bridges start, the flux of the cores is zero, while in steady state the flux vector of the three transformers runs on a hexagon centred in the origin, with the vertices at λ_{pk} (Figure 5). A plain start of the six-step pattern runs the same hexagon around a centre displaced by λ_{pk} from the origin: the magnetizing currents get a DC offset of up to $\lambda_{\mathrm{pk}}/L_{\mathrm{m}} = 13.3\,\mathrm{A}$, which decays with τ_{dc}. This is the offset seen in all the runs of Section 7 (2.1 A, −8.3 A and 6.2 A at 0.3 s in run 01).

The start that avoids it is the three-phase form of the half-width first pulse of the single-phase converters. From zero flux, the first state of the pattern is applied for a whole step: the flux vector moves along the direction of that state by $\frac{2}{3} V \cdot \frac{T_{\mathrm{s}}}{6} = \lambda_{\mathrm{pk}}$ and lands on a vertex of the centred hexagon. The pattern then continues with the state after the next one, skipping one state; from there on the flux runs on the centred hexagon. The transition that skips the state commutates two legs at once. The flux-centred start of Figure 5 was checked numerically on the ideal phase voltages (`py/three_phase_dab_figures.py`):

the mean flux over the second period is zero, against λ_{pk} for the plain start. The same rule applies after any stop of the bridges with the flux back at zero.

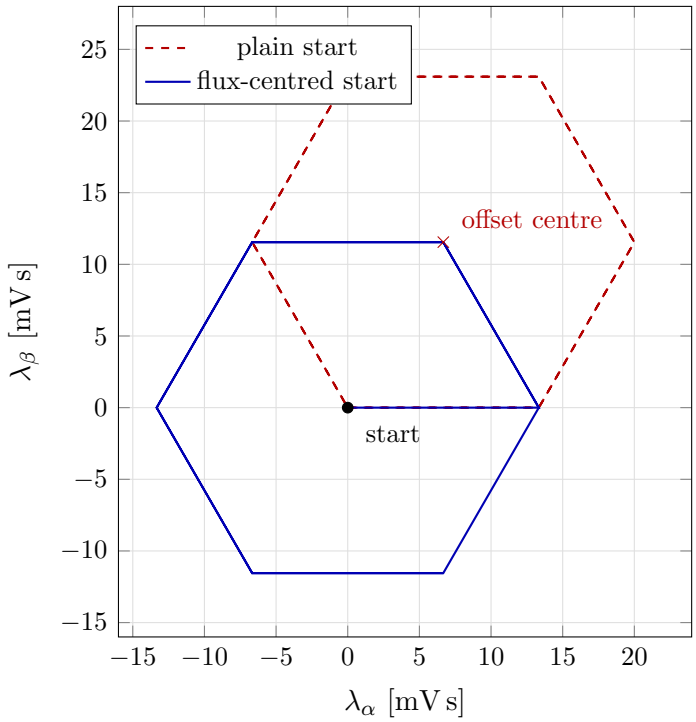


Figure 5: Space vector of the flux linkage of the three transformers over the first two periods, ideal six-step phase voltages at 1200 V, 10 kHz (`figures/flux_start.dat`). The plain start runs the steady-state hexagon around a centre displaced by $\lambda_{\mathrm{pk}} = 13.3\,\mathrm{mV\,s}$; the flux-centred start applies the first state for a whole step, skips the next state and runs the hexagon centred in the origin.

5.4. Updates of the phase shift

A change of φ moves the edges of bridge 2. If a leg sees the increment δ first on its rising edge, the low interval before that edge is longer by δ while the following high interval keeps its length: the leg loses $V\delta/(2\omega)$ of volt-seconds, with respect to the midpoint of the DC link, and the windings get a volt-second step, which appears as a DC offset of the currents [4]. A leg that sees the increment first on its falling edge gains the same amount. With a control that updates φ once per period at a fixed angle, the same legs see the increments first on the same edges period after period, and a ramp of φ at the rate $\dot{\varphi}$ becomes a DC voltage

$$V_{\mathrm{dc,leg}} = \frac{V}{2}\,\frac{\dot{\varphi}}{\omega} \tag{5.4}$$

on each leg, with signs that depend on the angle of the update. At the rate of the bench, $20\,\mathrm{p.u.\,s^{-1}}$ or $600\,^\circ\,\mathrm{s^{-1}}$, this is 0.1 V per leg and up to 0.13 V on a phase: during the ramp about 28 A of DC current circulate through the windings of both sides (time constant



$L/(R_1 + R_2') \simeq 6\,\mathrm{ms}$), and the magnetizing current drifts at about $67\,\mathrm{A\,s^{-1}}$, some 7 A at the end of a ramp of 100 ms. The rule that avoids it: the first edge of each leg after an update moves by $\delta/2$, the following edges by δ. Each leg then stretches one low and one high interval by $\delta/2$, and its volt-seconds stay balanced. An equivalent implementation computes the edge times from a φ interpolated within the period.

In the bench the control and the modulator update φ at every step of the pattern, six times per period, which spreads the increments of a ramp evenly over the rising and falling edges. The four runs of Section 7, with different ramps (reversals in opposite directions, light and partial load), end with the same magnetizing offsets within 0.6 A: the offset of the runs comes from the start, not from the ramps.

5.5. Measuring the DC component with slow sampling

In steady state the phase currents have half-wave symmetry: their AC part contains only odd harmonics, and $i(\theta + \pi) = -i(\theta)$ except for the DC component. Any pair of samples half a period apart therefore gives the DC component exactly, whatever the waveform:

$$I_{\mathrm{dc}} = \frac{i(\theta_0) + i(\theta_0 + \pi)}{2}. \tag{5.5}$$

The two samples need not be in the same period, which is what makes slow sampling possible:

- **Sampling period** $(m + \frac{1}{2})\,T_{\mathrm{s}}$**.** Consecutive samples fall at θ_0 and $\theta_0 + \pi$ of different periods: at $f_{\mathrm{s}} = 10\,\mathrm{kHz}$, 6.67 kHz ($m = 1$), 4 kHz ($m = 2$) or 2.86 kHz ($m = 3$), and the mean of two consecutive samples is I_{dc}. Sampling at an integer submultiple of f_{s} always falls at the same angle and cannot separate the DC component from the AC part.
- **Control at** f_{s}**.** A control that samples once per period can alternate the trigger of the ADC between θ_0 and $\theta_0 + \pi$ (the two event points of the timer of a leg), and pair consecutive samples.
- **Edge currents.** The currents at the rising and at the falling edge of the same leg are such a pair; they are also the switching currents, so the same samples monitor the ZVS. The analysis of the runs uses exactly this to separate the start offset from the switching currents (Section 6.3).
- **Without synchronous sampling.** An isolated sigma-delta modulator with a sinc filter whose window is an integer number of periods has zeros at f_{s} and at all its harmonics: its output, at any rate, is the period average, that is I_{dc}. Analog peak detectors of the positive and of the negative peak, read by a slow ADC, give $I_{\mathrm{dc}} = (\hat{I}_+ - \left|\hat{I}_-\right|)/2$.

The offset of the current sensor adds directly to the measured I_{dc} and becomes a true



DC current once the loop is closed: a Hall sensor with an offset of 0.5 % of a 500 A range is 2.5 A. The sensors need an auto-zero with the converter stopped, or a closed-loop or fluxgate type. A direct measurement of the flux, with a small sensor in the magnetic circuit, is an alternative to the measurement of the currents [5].

5.6. Cancelling it

With floating star points the three DC components of a side sum to zero: two regulators per side, one for each independent component (the α and β components of the DC vector, as in [6]), are enough. Each regulator trims the duty of the legs, moving the falling edge with respect to the rising one, and the DC voltage it produces on a leg is $V\,\Delta t\,f_\mathrm{s}$ for a trim Δt. With $R_1 = 2.34\,\mathrm{m\Omega}$, 1 ns of trim moves 3.4 A in steady state: the PWM needs a resolution well below a nanosecond, which the high-resolution PWM units of the controllers provide (about 150 ps, 0.5 A).

Each bridge acts on the DC current of its own side, and the two sides are coupled through the transformer. With $R_1 = R'_2 = R$ and the leakage neglected (below about 27 Hz), the DC components of a phase obey

$$i_1 = \frac{v_1 - v_2}{2R} + \frac{i_\mathrm{m}}{2}, \qquad i'_2 = \frac{v_1 - v_2}{2R} - \frac{i_\mathrm{m}}{2}, \qquad \tau_\mathrm{dc}\,\frac{\mathrm{d}i_\mathrm{m}}{\mathrm{d}t} = -i_\mathrm{m} + \frac{v_1 + v_2}{R}, \tag{5.6}$$

where v_1 and v_2 are the DC voltages that the two bridges put on the phase (side 2 referred to side 1), i'_2 is the DC current into bridge 2 referred to side 1, and i_m is the DC magnetizing current. The difference $v_1 - v_2$ drives a current that circulates through the windings of both sides and responds at once, through $2R$; the sum $v_1 + v_2$ drives the magnetizing current through the lag of Eq. (5.2). A regulator on each side, fed with the DC current of that side, controls both modes: the circulating mode is a resistive plant, the magnetizing mode a first-order lag with its pole at $1/(2\pi\tau_\mathrm{dc}) = 0.19\,\mathrm{Hz}$. A pure integrator with a bandwidth above that pole sees the magnetizing branch as a second integrator and is poorly damped (about 5° of phase margin at 2 Hz). The regulators are therefore PI: the proportional gain k_p acts as a virtual resistance in series with the phase and shortens the time constant of the magnetizing mode to $\tau_\mathrm{dc}/(1 + k_\mathrm{p}/R)$, 0.09 s with $k_\mathrm{p} = 20\,\mathrm{m\Omega}$, and the integral removes the static error left by the asymmetries. The note [6] reaches the same conclusion for transformers whose L_m/R is about 100 s. A bandwidth of 1 to 5 Hz is enough, since the disturbances (thermal drift of the delays and of the on-resistances) are slow, and cancelling the DC currents of both sides also cancels the flux offset. A series capacitor in the phases of one side would block the steady DC current without a regulator, but it has to carry 161 A rms and it resonates with L_m at a few hundred hertz, which needs damping.



6. Simulation model in advanced-dcdc-converters

6.1. The bench and its parameters

The folder analysis_three_phase_dab of [9] holds the model three_phase_dab.slx, built with the blocks of the library [10]. The root of the model has battery_1 and battery_2 (library block lithium_ion_battery on side 1 and side 2), the converter dab_modA, the global timing block, the measurment block, which logs every *_sim variable at the solver step in the last param.time.t_measure seconds of the run, and the scopes. Inside dab_modA, the subsystem dcdc_with_galvanic_isolation has the DC chokes (two per side, half of L_{dc} each), the DC links CFi_dc1 and CFi_dc2, the two bridges three_phase_inverter_mosfet_based_with_thermal_model with the device dataset and the heatsink of the parameters, and the three single_phase_transformer blocks; the subsystem dab_ctrl has the modulator three_phase_dab_modulator, the normalization of the feedbacks and the control of Section 4, triggered once per step of the pattern.

The model reads only the structure param, built by init_model.m with one group per subsystem (Table 9). The numbers of this converter are in private/dab3_param.m; the library provides the setup classes and functions, used as they are, and private/library_setup.m puts the library on the path (the copy already on the path, the folder library next to the repository, or a clone from GitHub). The bench case is selected with param.sim.bench_case (Section 7.1).

Table 9: Parameter structure of the bench (`init_model.m`).

Group	Content
`param.sim`	bench case, run length (0.3 s), solver step t_c (100 ns), iterations of the nonlinear solver
`param.time`	inputs of the shared timing object: base PWM frequency of the other converters of the template, logged window (20 ms), dead times
`param.glb_time`	timing object of the library (`timing_setup`): f_{PWM} of the DAB3, sample time of its control, solver step, triggers
`param.dab3`	the DAB3: `pwr_nom`, `fs`, `pwm`, `hw_in` (inputs of `hw_dab_three_phase_setup`) and `hw` (its output, with the series inductance of the sizing written over), `tank`, `mod` (φ_{max}, k_{ff}), `ctrl` (`ctrl_dab_setup`), `mosfet` (`device_mosfet_setup`), `bench`, `ic` (initial voltages of the DC links)
`param.battery`	`pack_1`, `pack_2` (`lithium_ion_battery_setup`, R_0 and R_1 halved and C_1 divided by 200 for a stiffer battery), open-circuit voltages at the initial SoC
`param.meas`	quantization of the measurements (11 bits)
`param.thermal`	heatsink of the bridges (`liquid_cooled_plate_2kw_setup`)
`param.dev`	names of the device datasets

6.2. Changes from the original model

The original model of the folder was built from a template shared with other converters: one initialization for active front end, inverters, CLLC, phase-shifted full bridge and DAB, with the 690 V application case of the library data. Table 10 lists what the review found and what was changed.



Table 10: Changes from the original model.

Item	Original model	Now
Transformer frequency	$f_{\mathrm{PWM}} = 3 \times 4\,\mathrm{kHz} = 12\,\mathrm{kHz}$; the modulator divides by 6: 2 kHz on the transformers	$f_{\mathrm{PWM}} = 6 f_{\mathrm{s}} = 60\,\mathrm{kHz}$: 10 kHz
Data	library classes of the template: DAB3 data at $3 \times 250\,\mathrm{kW}$; DC links, DC chokes, batteries and reference of the voltage loop from the single-phase DAB data	`private/dab3_param.m`: 250 kW, the numbers of this converter; library classes and blocks used as they are
Series inductance	derived by the class from f_{PWM} with the single-phase expression	28 µH from Eq. (3.1) at 30°, written over the value of the class
Modulator limit	$\varphi_{\max} = 2\pi/3$, beyond the maximum of the power at $\pi/2$: the gain changed sign inside the range of u	$\varphi_{\max} = \pi/3$
Feed-forward	$k_{\mathrm{ff}} = 0.25$ (30° at 1 pu with $2\pi/3$)	$k_{\mathrm{ff}} = 0.5$ (30° at 1 pu with $\pi/3$)
PI gains	$k_{\mathrm{p}} = 0.02$, $k_{\mathrm{i}} = 1$: the error of the feed-forward lasted about 1 s	$k_{\mathrm{p}} = 0.01$, $k_{\mathrm{i}} = 60$: 18 Hz, 90°, 14 dB
Control trigger	double update: the PI integrated with half its true sample time	single update, sample time $1/f_{\mathrm{PWM}}$
Dead time	2 µs	500 ns; in both cases not applied by the library modulator
Bench	fixed steps of the reference	four cases (Section 7.1); DC links and batteries start at the open-circuit voltage

6.3. Numerical settings and artefacts

The Simscape network runs on a local solver with the Backward Euler method [11] and the fixed step $t_{\mathrm{c}} = 100\,\mathrm{ns}$, 1000 steps per period; the Simulink solver is `ode23tb` with the maximum step t_{c}. The run lasts 0.3 s and the last 20 ms are logged at every step, which is what the loss analysis needs. The script `dab3_run_check.m` compares a logged window with the sizing; it accounts for the following artefacts of the model.

- **Numerical damping.** Backward Euler dissipates $\frac{1}{2}L(\Delta i)^2$ per step in every inductance, that is on average

$$P_{\mathrm{num}} = \frac{t_{\mathrm{c}}}{2L}\,\overline{v_{\mathrm{L}}^2} \tag{6.1}$$

  for an inductance L with the voltage v_{L}. The leakage inductances carry currents



with slopes up to $30\,\mathrm{A\,\mu s^{-1}}$: in run 01 the numerical damping is 0.87 kW in L_{d1} and L_{d2} and 0.05 kW in L_{m}, more than the physical losses of the model (windings 0.20 kW, cores 0.30 kW, conduction 0.16 kW). It scales with t_{c} (0.45 kW at 50 ns). `dab3_run_check.m` computes it from the logged currents and gives the efficiency without it.

- **Resolution of the phase shift.** The modulator rounds the edges to the solver step: φ moves in steps of $360° f_{\mathrm{s}} t_{\mathrm{c}} = 0.36°$, about 1 % of the power at 28°. The current loop toggles φ between two steps and the toggling excites the LC of the DC side: a limit cycle of ±3 A at about 500 Hz on 208 A in run 01 (Figure 7), ±20 % of the current at 0.1 pu. It is the resolution of the model: a digital PWM with 1 ns to 10 ns of resolution gives 0.004° to 0.04°.
- **Switch model.** The bridges are ideal switches with a thermal network. They conduct 13 mA in the off state at 1.2 kV, about 0.1 kW for the 12 switches. Their `ploss` outputs add E_{on} and E_{off} of the dataset at every edge, as in hard switching (2.7 kW in run 01), while the electrical circuit has no switching loss. There is no dead time and no C_{oss} (Section 3.4).
- **Flux offset of the start.** The flux of the transformers starts at zero (Section 5.3); the magnetizing currents keep an offset of several amperes at the end of the run. It shifts the switching currents of the rising and of the falling edges of a leg in opposite directions: `dab3_run_check.m` averages the two edges, by Eq. (5.5), and reads the offset from the magnetizing current.
- **Batteries.** They start at SoC 0.85 (about 1248 V) and move by 20 V to 40 V at 208 A: $d \simeq 1 + 0.05$ times the load in forward power, $1 - 0.05$ times the load in reverse power.

7. Runs

7.1. Bench cases

The bench cases of `param.sim.bench_case` share the same sequence: the reference is zero until 10 ms, goes to a first value with the ramp of $20\,\mathrm{p.u.\,s^{-1}}$, and at 0.1 s to a second value, which lasts to the end of the run and includes the logged window, 0.28 s to 0.30 s (Table 11). The reversals of the power happen before the logged window: they can be watched on the scopes, or logged with a longer window and a decimation, at the cost of the loss analysis, which needs every step.

Table 11: Bench cases (`param.sim.bench_case`, numbers in `param.dab3.bench`).

Case	First value	From 0.1 s	Purpose	Run
`forward`	−1 pu	1 pu	nominal point, Sections 2 and 3	01
`reverse`	1 pu	−1 pu	the same quantities with the power from side 2 to side 1	02
`light`	−1 pu	0.1 pu	switching currents set by the magnetizing current	03
`mismatch`	−1 pu	0.25 pu	battery 2 at SoC 0.3 (1184 V open circuit): ZVS limit of bridge 2	04

7.2. Results

Table 12 gives the results of the four runs (`sim_results_01.mat` to `sim_results_04.mat`, $t_c = 100$ ns, window 0.28 s to 0.30 s), with the harmonic model at the measured V_1, V_2 and φ in brackets. The phase shift is the mean over all the edges of the window, the six legs and both directions, because the edges of a leg move by one step from period to period (the period of the modulator clock is not an integer number of solver steps) and the limit cycle moves φ by one step.

Table 12: Results of the four runs; in brackets the harmonic model at the measured voltages and phase shift.

Quantity	01 `forward`	02 `reverse`	03 `light`	04 `mismatch`
V_1 / V_2	1210 V / 1269 V ($d = 1.05$)	1271 V / 1208 V ($d = 0.95$)	1237 V / 1243 V ($d = 1.005$)	1233 V / 1186 V ($d = 0.96$)
Power, side 1 and side 2	252.1 kW in, 250.4 kW out	264.7 kW out, 266.5 kW in	25.7 kW in, 25.1 kW out	64.3 kW in, 63.6 kW out
DC input current of side 1	208.4 A (reference 208.3 A)	−208.3 A	20.8 A	52.2 A
Phase shift	28.25°	−30.14°	2.56°	6.83°
Phase current, rms	157.7 A (157.6)	168.5 A (168.4)	16.4 A (15.1)	42.1 A (41.6)
Switching current, bridge 1	99.1 A (100.5)	151.8 A (151.8)	13.4 A (14.8)	50.2 A (51.6)
Switching current, bridge 2	143.1 A (143.0)	105.7 A (107.1)	19.6 A (19.6)	16.0 A (15.9)
Transition, bridge 1 / 2 (Eq. (3.3), dead time 500 ns)	80 ns / 58 ns	55 ns / 75 ns	607 ns / 415 ns	161 ns / 487 ns
Losses between the DC links	1.66 kW	1.76 kW	0.57 kW	0.68 kW
of which numerical damping, off-state conduction	0.91 kW, 0.10 kW	0.97 kW, 0.10 kW	0.13 kW, 0.10 kW	0.25 kW, 0.10 kW
windings, cores, conduction	0.20, 0.30, 0.16 kW	0.22, 0.29, 0.18 kW	0.00, 0.33, 0.00 kW	0.01, 0.31, 0.01 kW
Efficiency without numerical damping and off-state conduction	99.74 %	99.74 %	98.69 %	99.48 %
Limit cycle of I_{dc1} (about 500 Hz)	±3 A	±0.6 A	±4 A	±3.4 A
Magnetizing current offset, phases a, b, c	2.1, −8.3, 6.2 A	1.6, −8.5, 6.8 A	1.7, −8.2, 6.5 A	1.6, −8.1, 6.5 A

7.3. Run 01, forward power at 1 pu

The loop holds 208.4 A against a reference of 208.3 A. The batteries have moved to 1210 V and 1269 V, so the run is at $d = 1.05$ and $\varphi = 28.25°$ rather than at the nominal 30°. Figure 6 compares one period of the phase currents and of the magnetizing current with the harmonic model at the same point: the difference is 1.5 A rms on 158 A, and the switching currents agree within 1.5 A. With $d > 1$ bridge 1, at the lower voltage, commutates less current than bridge 2 (99 A against 143 A), both well above the 16 A of Eq. (3.3). The run loses about 1 kW more than the harmonic model: the numerical damping and the off-state conduction of the switch model. Without them the efficiency between the DC links is 99.74 %; with the turn-off estimate of Section 3.6 at this point (1.5 kW, upper bound) the total is about 2.2 kW, $\eta \simeq 99.1\,\%$, as the harmonic model gives for $d = 1.05$ at full load. Figure 7 shows the limit cycle of the phase quantization over the logged window.

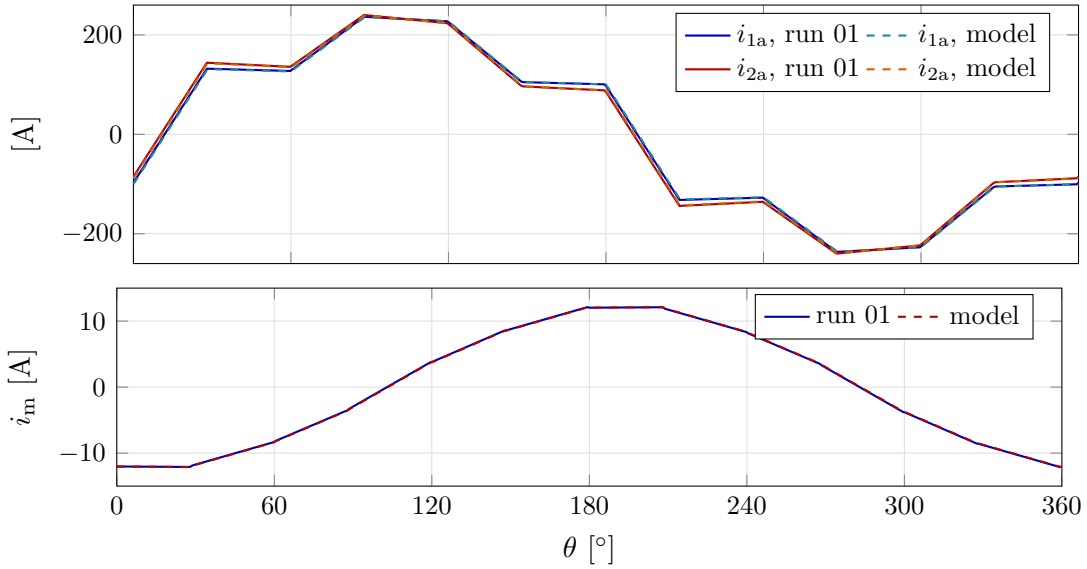


Figure 6: Run 01: phase currents of phase a on both sides (top) and magnetizing current $i_{1a} - i_{2a}$ (bottom), averaged over the 199 periods of the window, aligned on the rising edge of leg a of bridge 1, with the DC offset of the window removed; dashed, the harmonic model at 1210 V, 1269 V, 28.25° (`figures/run01_period.dat`, `py/three_phase_dab_runs.py`).

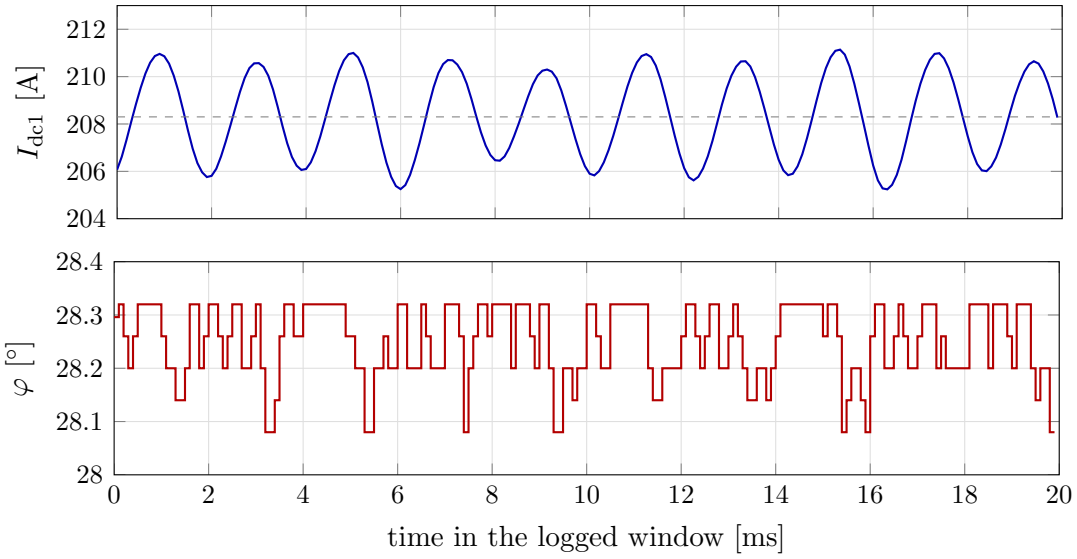


Figure 7: Run 01, logged window: DC input current of side 1 (top, reference dashed) and phase shift (bottom), both averaged over each period (`figures/run01_dc.dat`). The edges move in steps of 0.36°; averaged over the six edges of a period the phase shift moves in steps of 0.06°, and its toggling excites the LC of the DC side at about 500 Hz.

7.4. Run 02, reverse power

With the reference at −1 pu the power flows from side 2 to side 1 and the bridges exchange their roles: bridge 2 drives, bridge 1 rectifies, and with battery 1 charging at 1271 V the run is at $d = 0.95$, the mirror image of run 01. The currents and the switching currents agree with the model within 1.4 A. Because the loop regulates the current of side 1, the power is 266.5 kW on side 2 (Section 4.4).

7.5. Run 03, light load

At 0.1 pu the switching currents are set by the magnetizing current (Figure 8). Bridge 1, at the lower voltage ($d = 1.005$), commutates 13.4 A against the 16.2 A that Eq. (3.3) needs at 1237 V: the leg voltage reaches 80 % to 90 % of its swing before the turn-on, a partial ZVS with about 10 W of residual turn-on loss. The map of Figure 4 puts the limit at about 11 % of the load at this d, and the run is just below it. The cores dominate the losses (334 W, 1.3 % of 25 kW), with R_{fe} still a placeholder. One step of φ is 14 % of the power at 2.5°, so the limit cycle is ±20 % of the current. The flux offset of the start, 8 A in phase b, brings some edges of bridge 1 to a negative switching current (minimum −1.6 A): in the converter, the flux-centred start of Section 5.3 avoids it.

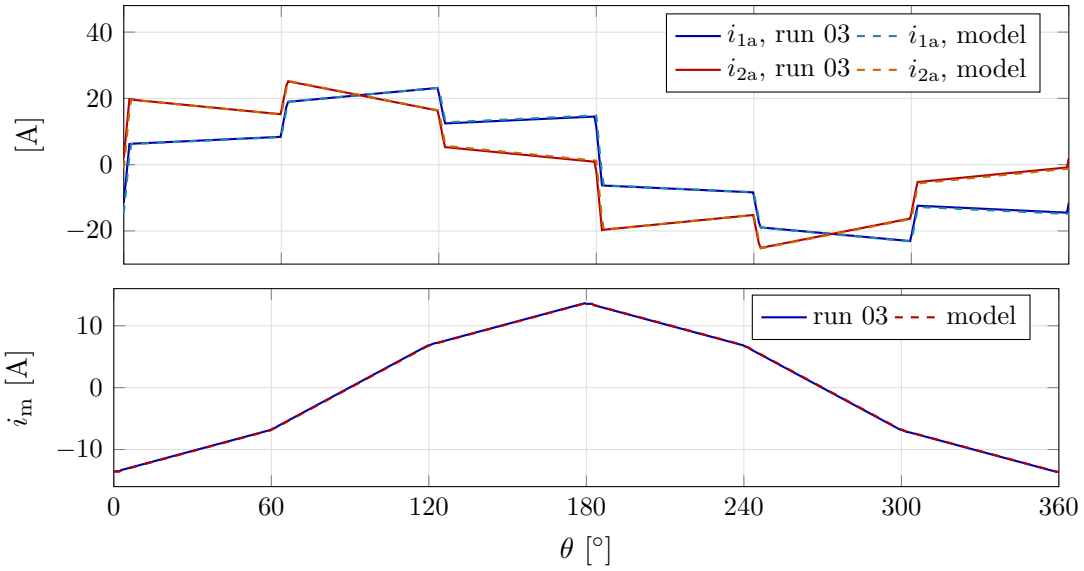


Figure 8: Run 03, light load, as Figure 6; harmonic model at 1237 V, 1243 V, 2.56° (`figures/run03_period.dat`). The phase currents are almost the magnetizing current, which sets the switching currents.

7.6. Run 04, voltage mismatch

With battery 2 at SoC 0.3 and 0.25 pu the run is at $d = 0.96$. Bridge 2, at the lower voltage, commutates 16.0 A (model 15.9 A) against the 15.6 A needed in 500 ns at 1186 V: the point sits on the ZVS boundary of Figure 4, as the map predicts (25 % of the load at $d = 0.96$). Bridge 1 commutates 50 A. With the flux offset of the start the edges of bridge 2 spread from −1.2 A to 20 A.

7.7. Summary of the runs

The four runs confirm the harmonic model, with the currents and the switching currents within 2 % and the power within 0.5 % once the numerical damping is accounted for, and they confirm the ZVS map in both directions of the power, at light load and with different battery voltages. The model artefacts are known and quantified: numerical damping, off-state conduction, phase quantization and the flux offset of the start.

8. Open points

- **Specification.** DC voltages and their range on both sides, and the transformer frequency: 1200 V / 1200 V and 10 kHz are working assumptions. The voltage range decides whether the ZVS range of Table 6 is enough.
- **Turn-off energy.** The soft turn-off energy of the module at 1200 V and 100 A to 150 A decides between 99.0 % and 99.6 % of efficiency; a module of the 300 A to



600 A class is to be evaluated.

- **Transformers.** The leakage of 12 µH, the winding resistance, the gap for 1 mH and the core losses (R_{fe} is a placeholder that dominates the light-load losses) need the design of the windings; the external inductor of about 8 µH per side is to be designed.
- **Transitions in the model.** The library modulator does not apply the dead time and the bridges have no C_{oss}; the ZVS transitions, the partial ZVS of run 03 and the self-balancing of Section 5.2 need a bridge with C_{oss} and a modulator with dead time.
- **Control at 10 kHz.** The PI of the model runs at 60 kHz; it should be moved to one update per period, with the margins checked again (Section 4.3) and the update rule of Section 5.4.
- **DC bias.** Flux-centred start, update rule of the phase shift and DC regulators (Section 5) are to be implemented and tested in the model, with deliberate asymmetries of the gate timing.
- **Reverse power.** A power reference, or the current reference scaled by the battery voltage, to keep 250 kW in both directions.
- **Numerics.** The numerical damping is about 1 kW at 100 ns; efficiency figures need the correction of `dab3_run_check.m` or a step of 50 ns. The limit cycle is a feature of the phase resolution of the model.
- **DC side.** DC links and chokes (500 µF, 250 µH) are the values of the bench, to be confirmed with the batteries and the cabling.

9. Conclusions

The 250 kW isolated stage is a three-phase dual active bridge between two batteries at about 1200 V: two six-step bridges of 2 kV SiC modules, three single-phase transformers in star-star on nanocrystalline cut cores with 14 turns and a gapped magnetizing inductance of 1 mH, 28 µH of series inductance per phase for 30° at the nominal power, and a modulator limit of 60°, below the maximum of the power. At 10 kHz both bridges switch with ZVS at 124 A, down to 9 % of the load with equal voltages, and the losses are estimated at 2.4 kW with the turn-off term as an upper bound. Against a single-phase DAB at the same point the DAB3 carries two thirds of the current per winding, turns off less than half the current per commutation and has a DC-link ripple six times smaller. The current loop on the DC choke current, with a feed-forward of the reference, has 18 Hz of bandwidth, set by the LC of the DC side, and can run at the transformer frequency or slower. The DC bias of the transformers is a slow phenomenon (time constant up to 0.85 s) driven by tiny asymmetries (34 A for 10 ns): it is limited by the gap and by the



ZVS transitions, avoided at the start and in the updates of the phase shift by two modulation rules, and cancelled by slow regulators fed with sample pairs half a period apart. The Simscape bench of `advanced-dcdc-converters`, rebuilt around its own parameter structure with the blocks of the library, reproduces the harmonic model within 2 % in four runs, forward and reverse at 1 pu, light load and voltage mismatch, and its artefacts (numerical damping, phase quantization, off-state conduction, flux offset of the start) are known and quantified.

A. Files

Table 13: Files referred to in this note.

File	Content
`advanced-dcdc-converters`, folder `analysis_three_phase_dab`	
`three_phase_dab.slx`	Simulink/Simscape model of the bench, reads only `param.*`
`init_model.m`	builds `param`, writes the series inductance of the sizing over the library one, sets the initial voltages, opens the model
`private/dab3_param.m`	the numbers of the DAB3: operating point, PWM, hardware, series inductance, modulator, PI gains, bench cases
`private/library_setup.m`	puts the library on the path, cloning it when missing
`private/dab3_ideal_op.m`, `private/dab3_harmonic_op.m`	phase shift for a power (Eq. (2.2)) and harmonic model (Section 2.3)
`dab3_analysis.m`	steady-state analysis of the sizing: nominal point, power against φ, map over load and d, ZVS boundaries
`dab3_run_check.m`, `private/dab3_edges.m`, `private/dab3_isw.m`	check of a logged run against the sizing: powers, phase shift, switching currents, loss breakdown with the numerical damping, magnetizing offset, DC side
`dab3_sizing.md`, `README.md`	sizing note and description of the bench in short form
`scripts/plotting_sim_results.m`, `spectrum.m`	figures and spectrum of a run
`library`, blocks and functions used by the bench	
`three_phase_dab_modulator`	six-step modulator at $f_{\mathrm{PWM}}/6$, $\varphi = u\,\varphi_{\max}$



File	Content
three_phase_inverter_mosfet_based_with_thermal_model	the two bridges, with the dataset danfoss_SKM1700MB20R4S2I4
single_phase_transformer, lithium_ion_battery, pi_ctrl	transformers, batteries, PI
hw_dab_three_phase_setup, timing_setup, ctrl_dab_setup, device_mosfet_setup, lithium_ion_battery_setup	setup classes and functions
repositories-documentation, folder repos_doc/three_phase_dab	
three_phase_dab.tex, three_phase_dab.pdf	this note
py/three_phase_dab_figures.py	harmonic model and data of the analytic figures (standard library only); prints the numbers of Tables 1, 3 and 6
py/three_phase_dab_runs.py	data of the figures of the runs, from sim_results_01.mat and sim_results_03.mat (numpy, h5py)
figures/*.dat	data files of the figures

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[10] pwr-control, *library* repository (Simscape components, masked blocks and documentation), https://github.com/pwr-control/library.

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